

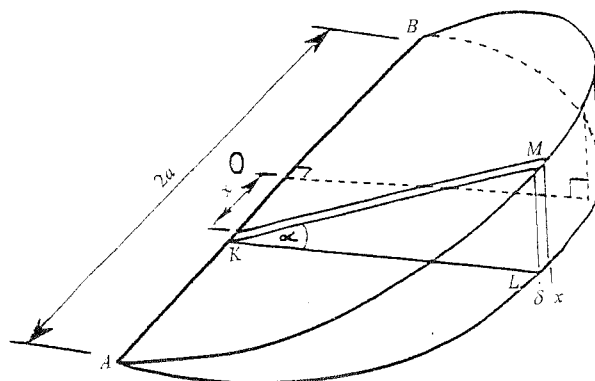
QUESTION 1 (15 Marks)

Marks

- (a) The base of a solid is the region in the first quadrant bounded by the curve $y = \sin x$, the x -axis and the line $x = \frac{\pi}{2}$. Cross-sections of this solid perpendicular to the x -axis are equilateral triangles with one side lying in the base of the solid. Find the exact volume of the solid. 6
- (b) A particle of mass m is projected upwards in a resistive medium where the force against the motion is inversely proportional to v , where v is the velocity of the particle, ie; $m\ddot{x} = -mg - \frac{mK}{v}$, where K is a constant and g the acceleration due to gravity. 9
- (i) If the initial velocity of projection is U m/s, show that the time taken by the particle as a function of its velocity is given by an equation of the form $t = A + B \ln C$, and find expressions for A, B and C .
- (ii) Derive an expression for the time taken to reach its maximum height in this medium.

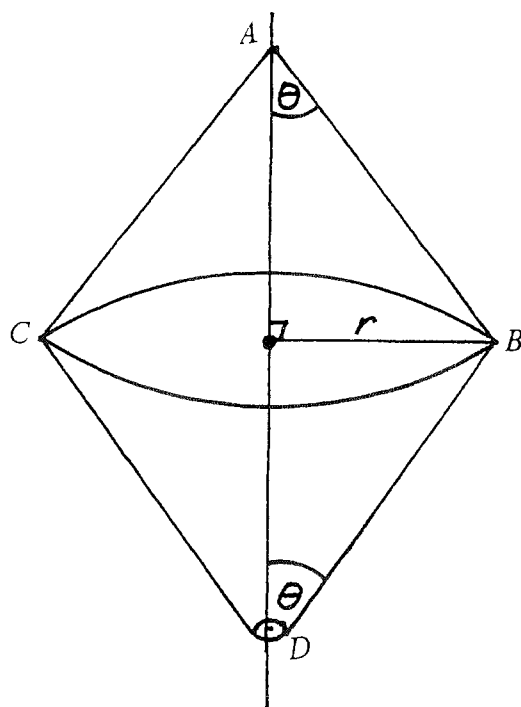
QUESTION 2 (15 Marks)

- (a) A hole of diameter a centimetres is bored through the centre of a solid sphere of diameter $2a$ centimetres. Use the "method of cylindrical shells" to find the exact volume of the remaining solid. 6
- (b) The solid wedge as shown was made by slicing a right cylinder of radius a at an angle α through diameter AB of its base. A triangular slice of thickness δx perpendicular to the base and line AB is positioned at distance x from the centre O as in the diagram. 9
- (i) Show that ML , the height of the triangular slice is $\sqrt{a^2 - x^2} \tan \alpha$.
- (ii) Deduce a formula for the volume of the wedge.
- (iii) Given that $\alpha = \frac{2\pi}{n}$ and $\tan \alpha$ decreases whilst n increases, find the volume of n identical wedges with a common diameter AB .



QUESTION 3 (15 Marks)

- (a) The acceleration of a body of unit mass moving towards earth under gravitational attraction varies inversely as the square of its distance from the centre of the earth, 5
 ie; $\ddot{x} = \frac{-k}{x^2}$ (k a constant). If the body starts from rest at a distance a from the centre of the earth, show that its speed at a distance x from the centre of the earth is $v = \sqrt{\frac{2k(a-x)}{ax}}$.
- (b) Two equal masses are connected to the ends of two rods AB and AC (in the same plane) of equal length which are hinged together at the point A to a vertical shaft. Two supporting rods DB and DC are also hinged together to a ring D which can slide up and down the shaft. The rods $AB = AC = DB = DC = L$, the masses at B and C are M kg and the ring has mass m kg. 10
- (i) Copy the diagram onto your answer sheet showing all the acting forces.
- (ii) Show that the semi-vertical angle θ when rotating at a speed of ω radians/second is given by $\sec \theta = \frac{ML\omega^2}{(M+m)g}$.

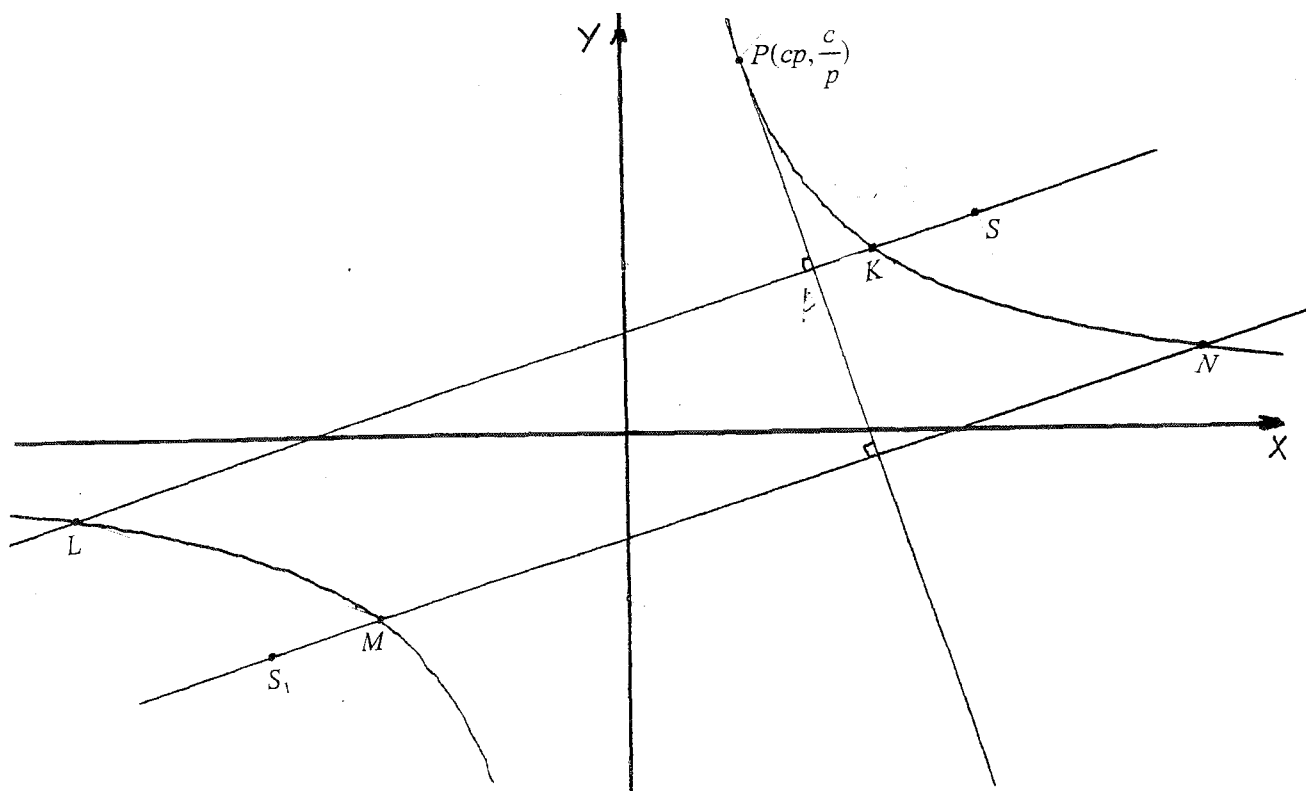


QUESTION 4 (15 Marks)

- (a) A square $ABCD$ has sides of length 1 unit. The asymptotes of hyperbola $xy = c^2$ lie on sides AB and AD of the square, and the point $C(1,1)$ is one of the foci. Show that the hyperbola bisects the other two sides. 3

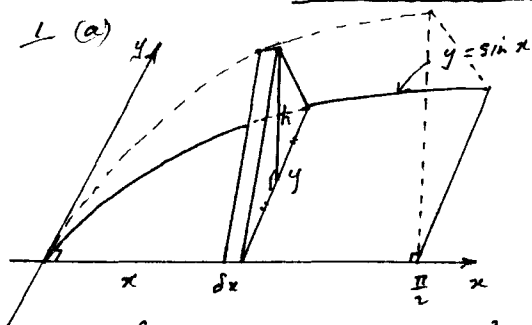
- (b) The perpendiculars drawn from the foci S and S_1 of hyperbola $xy = c^2$ to the tangent at a point $P(cp, \frac{c}{p})$ meet the curve at K, L, M, N as shown in the diagram. 12

- (i) Find the equation of the line SKL .
- (ii) Find, as a function of p , an expression for the parameter " k " at the point K .
- (iii) Hence, or otherwise, prove that $KLMN$ is a parallelogram.
- (iv) Show that the sides KN and LM of the parallelogram are perpendicular to the diameter through P .



END OF PAPER

SOLUTIONS TO
TERM 2 ASSESSMENT
EXTENSION II 2005



Area of triangular section is:

$$A = \frac{1}{2} y^2 \sin 60^\circ$$

$$= \frac{1}{2} \cdot y^2 \cdot \frac{\sqrt{3}}{2}$$

$$\therefore A = \frac{y^2 \sqrt{3}}{4}$$

$$\therefore dV = \frac{\sqrt{3}}{4} y^2 dx$$

Since $y = \sin x$ $\therefore y^2 = \sin^2 x$

$$\therefore V = \frac{\sqrt{3}}{4} \int_0^{\pi/2} y^2 dx = \frac{\sqrt{3}}{4} \int_0^{\pi/2} \sin^2 x dx$$

$$= \frac{\sqrt{3}}{8} \int_0^{\pi/2} (1 - \cos 2x) dx$$

$$= \frac{\sqrt{3}}{8} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/2}$$

$$= \frac{\sqrt{3}}{8} \left(\frac{\pi}{2} - 0 \right)$$

$$\therefore V = \frac{\pi \sqrt{3}}{16} U^3$$

(b)(i)

$$m\ddot{x} = -mg - \frac{mK}{v}$$

$$\therefore \ddot{x} = -g - \frac{K}{v}$$

$$\therefore \frac{dv}{dt} = -\left(\frac{gv+K}{v}\right)$$

$$\therefore \frac{v dv}{gv+K} = -dt$$

$$\therefore \int \frac{\frac{1}{2} (gv+K) - \frac{K}{g}}{gv+K} dv = -\int dt \text{ since } \frac{v}{gv+K} = \frac{\frac{1}{g}(gv+K) - \frac{K}{g}}{gv+K}$$

$$\therefore -\frac{1}{g} \int dv + \frac{K}{g} \int \frac{dv}{gv+K} = \int dt$$

$$\therefore -\frac{v}{g} + \frac{K}{g^2} \ln(gv+K) = t + C$$

When $t=0$, $v=U$

$$\therefore \frac{K}{g^2} \ln(gU+K) - \frac{U}{g} = C$$

$$\therefore -\frac{v}{g} + \frac{K}{g^2} \ln(gv+K) = t + \frac{K}{g^2} \ln(gU+K) - \frac{U}{g}$$

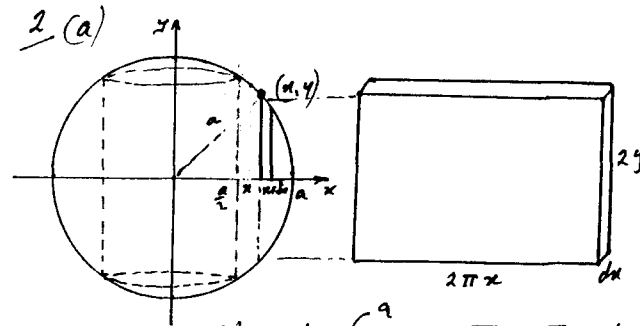
$$\therefore t = \left(\frac{U-v}{g}\right) + \frac{K}{g^2} \ln\left(\frac{gv+K}{gU+K}\right)$$

$$\therefore A = \frac{U-v}{g}, \quad B = \frac{K}{g^2}, \quad C = \frac{gv+K}{gU+K}$$

(ii) Maximum height occurs when $v=0$

$$\therefore t = \frac{U}{g} + \frac{K}{g^2} \ln\left(\frac{K}{gU+K}\right)$$

2 (a)



Let the volume of a cylindrical shell be:

$$dV = 2\pi x \cdot 2y \cdot dx$$

Since $y = \sqrt{a^2 - x^2}$

$$\therefore dV = 4\pi x \sqrt{a^2 - x^2} dx$$

$$\text{Now } V = \int_{\frac{a}{2}}^a 4\pi x \sqrt{a^2 - x^2} dx$$

$$= -2\pi \int_{\frac{3a^2}{4}}^0 u^{\frac{1}{2}} du$$

$$= 2\pi \int_0^{\frac{3a^2}{4}} u^{\frac{1}{2}} du$$

$$= 2\pi \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^{\frac{3a^2}{4}}$$

$$= \frac{4\pi}{3} \left[\left(\frac{3}{4} a^2 \right)^{\frac{3}{2}} - 0 \right]$$

$$= \frac{4\pi}{3} \cdot \frac{3\sqrt{3}}{8} a^3$$

$$\therefore V = \frac{\sqrt{3}}{2} \pi a^3$$

Let $u = a^2 - x^2$

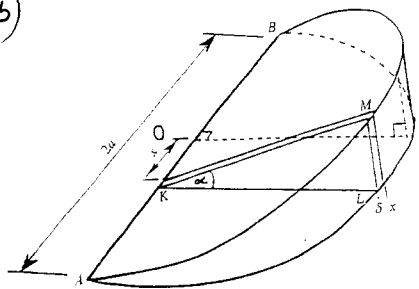
$$\therefore du = -2x dx$$

When $x = \frac{a}{2}$ $u = \frac{3a^2}{4}$

When $x = a$ $u = 0$

and $x = \sqrt{a^2 - u}$

2(b)



(i) From the diagram
 $KL^2 = OL^2 - OK^2$ by Pythagoras.

$$\therefore KL^2 = a^2 - x^2$$

$$\therefore KL = \sqrt{a^2 - x^2}$$

Now $\tan \alpha = \frac{ML}{KL}$

$$\therefore ML = KL \tan \alpha$$

$$\therefore ML = \sqrt{a^2 - x^2} \tan \alpha$$

(ii) Area of $\triangle KLM$ is $\frac{1}{2} \cdot KL \cdot LM$

$$\therefore A = \frac{1}{2} \cdot \sqrt{a^2 - x^2} \cdot \sqrt{a^2 - x^2} \tan \alpha$$

$$\therefore A = \frac{1}{2} (a^2 - x^2) \tan \alpha$$

$$\therefore \delta V = \frac{1}{2} (a^2 - x^2) \tan \alpha \delta x$$

$$\therefore V = \frac{\tan \alpha}{2} \times 2 \int_0^a (a^2 - x^2) dx$$

$$= \tan \alpha \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \tan \alpha \left(a^3 - \frac{a^3}{3} \right)$$

$$\therefore V = \frac{2}{3} a^3 \tan \alpha$$

(iii) as $\tan \alpha$ decreases,

$$\therefore \tan \alpha \rightarrow \alpha$$

$$\therefore V \rightarrow \frac{2a^3}{3} \alpha$$

For n identical wedges
 we have:

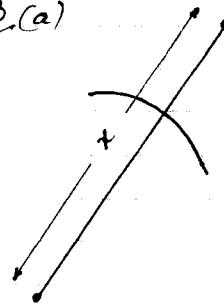
$$V_n = n \cdot \frac{2a^3}{3} \alpha$$

$$\text{For } \alpha = \frac{2\pi}{n}$$

$$\therefore V_n = n \cdot \frac{2a^3}{3} \cdot \frac{2\pi}{n}$$

$$\therefore V_n = \frac{4}{3} \pi a^3$$

3(a)



$$\ddot{x} = -\frac{\mu}{x^2} \therefore v \frac{dv}{dx} = -\frac{\mu}{x^2}$$

$$\therefore v dv = -\frac{\mu}{x^2} dx$$

$$\therefore \int v dv = -\mu \int \frac{dx}{x^2}$$

$$\therefore \frac{1}{2} v^2 = \frac{\mu}{x} + C$$

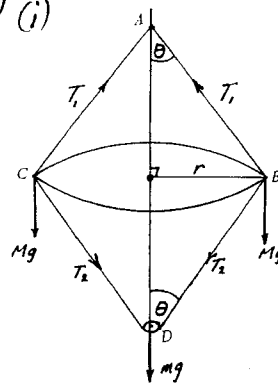
when $x = a, v = 0 \therefore C = -\frac{\mu}{a}$

$$\therefore \frac{1}{2} v^2 = \frac{\mu}{x} - \frac{\mu}{a}$$

$$\therefore v^2 = 2\mu \left(\frac{1}{x} - \frac{1}{a} \right) = 2\mu \left(\frac{a-x}{ax} \right)$$

$$\therefore v = \sqrt{\frac{2\mu(a-x)}{ax}} \text{ for } v > 0$$

(b) (i)



(ii) Resolving forces at B:

Vertically: $(T_1 - T_2) \cos \theta = Mg$ — (1)

Horizontally: $(T_1 + T_2) \sin \theta = Mr\omega^2$ — (2)

Forces at D: $2T_2 \cos \theta = mg$ — (3)

Now $T_2 = \frac{mg}{2} \sec \theta$ from (3)

Sub. into (1): $\left(T_1 - \frac{mg}{2} \sec \theta \right) \cos \theta = Mg$

$$\therefore T_1 = Mg \sec \theta + \frac{mg}{2} \sec \theta$$

$$= \left(Mg + \frac{mg}{2} \right) \sec \theta \text{ — (4)}$$

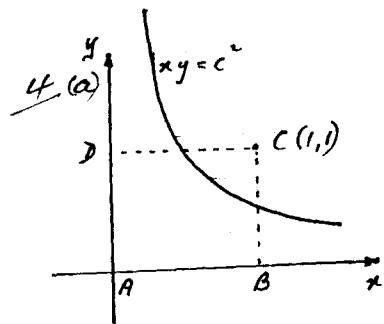
Sub. (4) into (2): $\left[\left(Mg + \frac{mg}{2} \right) \sec \theta + \left(\frac{mg}{2} \sec \theta \right) \right] \sin \theta = Mr\omega^2$

Since $\sin \theta = \frac{r}{l} \therefore \left[\left(Mg + \frac{mg}{2} \right) \sec \theta + \frac{mg}{2} \sec \theta \right] \sin \theta = M\omega^2 l \sin \theta$

$$\therefore r = l \sin \theta$$

$$\therefore \left(Mg + \frac{mg}{2} \right) \sec \theta = M\omega^2 l$$

$$\therefore \sec \theta = \frac{M\omega^2 l}{(M + \frac{m}{2})g}$$

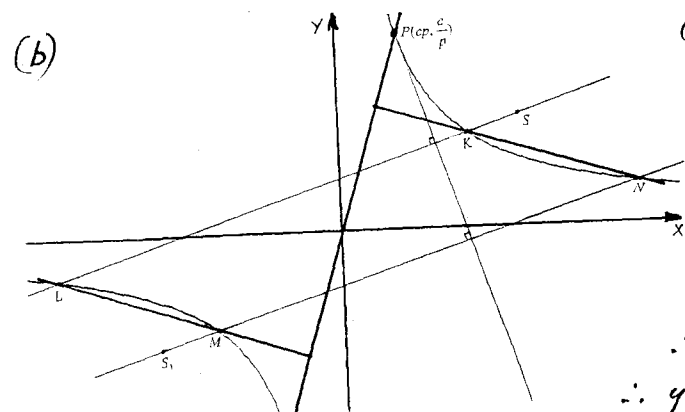


Now $C(1,1) \equiv C(c\sqrt{2}, c\sqrt{2})$
 $\therefore x = y = c\sqrt{2} \quad \therefore c\sqrt{2} = 1$
 $\therefore c^2 = \frac{1}{2}$

Since $xy = c^2$
 $\therefore xy = \frac{1}{2}$
 when $x=1, y = \frac{1}{2}$
 why $y=1, x = \frac{1}{2}$

$\therefore (1, \frac{1}{2})$ is mid-point of BC, and $(\frac{1}{2}, 1)$ mid-point of CD.

(b)



(i) Let equation be
 $y - y_1 = m(x - x_1)$
 $\therefore y - c\sqrt{2} = m(x - c\sqrt{2})$ at P
 $\frac{dy}{dx} = -\frac{c^2}{x^2}$
 $= -\frac{1}{p^2}$ at $x = cp$
 $\therefore m = -\frac{1}{p^2}$
 $\therefore y - c\sqrt{2} = -\frac{1}{p^2}(x - c\sqrt{2})$
 $\therefore y - c\sqrt{2} = -\frac{1}{p^2}x + \frac{c\sqrt{2}}{p^2}$
 $\therefore y - \frac{1}{p^2}x = c\sqrt{2}(1 - \frac{1}{p^2})$ — (1)

(ii) Let $K(x, y) \equiv K(ck, \frac{c}{k})$

Since K lies on (1)

$\therefore \frac{c}{k} - \frac{1}{p^2}ck = c\sqrt{2}(1 - \frac{1}{p^2})$

$\therefore \frac{1}{k} - \frac{1}{p^2}k = \sqrt{2}(1 - \frac{1}{p^2})$

$\therefore p^2k^2 + \sqrt{2}k - \sqrt{2}p^2k - 1 = 0$

$\therefore p^2k^2 + \sqrt{2}(1 - p^2)k - 1 = 0$

$\therefore k = \frac{\sqrt{2}(p^2 - 1) \pm \sqrt{[\sqrt{2}(1 - p^2)]^2 + 4p^2}}{2p^2}$

$= \frac{\sqrt{2}(p^2 - 1) \pm \sqrt{2(1 + p^4)}}{2p^2}$

$\therefore k = \frac{\sqrt{2}[(p^2 - 1) \pm \sqrt{1 + p^4}]}{2p^2}$

$\therefore k = \frac{\sqrt{2}[(p^2 - 1) + \sqrt{1 + p^4}]}{2p^2}$ for k in first quadrant.

(iii) Since L lies on SKL, then $l = \frac{\sqrt{2}[(p^2 - 1) - \sqrt{1 + p^4}]}{2p^2}$
 Similarly, for M and N which lie on the line through S, $(-c\sqrt{2}, -c\sqrt{2})$, the parameters m and n are $m = \frac{-\sqrt{2}[(p^2 - 1) + \sqrt{1 + p^4}]}{2p^2}$ and $n = \frac{-\sqrt{2}[(p^2 - 1) - \sqrt{1 + p^4}]}{2p^2}$

$\therefore m = -k$ and $l = -n$ i.e. diagonals of KLMN bisect each other at origin. $\therefore$ KLMN is a parallelogram.

(iv) The gradient of diameter OP is $m_{OP} = \frac{(\frac{c}{p})}{cp} = \frac{1}{p^2}$
 The gradient of KN is :

$m_{KN} = \frac{(\frac{c}{k} - \frac{c}{n})}{(ck - cn)}$ for $K(ck, \frac{c}{k})$ and $N(cn, \frac{c}{n})$
 $= \frac{c(n - k)}{c(k - n)kn}$

$= -\frac{1}{kn}$
 $= -1 \div \left\{ \frac{[\sqrt{2}(p^2 - 1) + \sqrt{1 + p^4}]}{2p^2} \times \frac{[-\sqrt{2}(p^2 - 1) - \sqrt{1 + p^4}]}{2p^2} \right\}$

$= -1 \div \left\{ \frac{2}{4p^4} [\sqrt{1 + p^4} + (p^2 - 1)][\sqrt{1 + p^4} - (p^2 - 1)] \right\}$

$= -1 \div \left\{ \frac{1}{2p^4} [(1 + p^4) - (p^4 - 2p^2 + 1)] \right\}$

$= -1 \div \left[\frac{1}{2p^4} (1 + p^4 - p^4 + 2p^2 - 1) \right]$

$= -1 \div \frac{2p^2}{2p^4}$
 $= -p^2$

Now $m_{OP} \times m_{KN} = \frac{1}{p^2} \times -p^2 = -1$

$\therefore KN \perp OP$

Since $KN \parallel LM$ (opposite sides of parm KLMN parallel)
 $\therefore LM \perp OP$